

PROJECT PROPOSAL: GDC-CMS

**GDC-CMS: A CONTENT MANAGEMENT SYSTEM FOR
INSTITUTIONAL INFORMATION DISSEMINATION**



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Project Overview

The GDC Digital Portal is a modern website designed specifically for Government Degree College (GDC) Gulabad. Currently, most college news is shared on physical notice boards or simple websites that are hard to update. This project changes that by creating a "live" system where the college can share news, teacher information, and photos instantly.

Problem Statement

Currently, college information systems often rely on manual updates or static websites that require technical expertise to modify. This leads to:

Delayed communication of urgent notices (Exams/Admissions).

Outdated faculty records and department information.

Lack of a centralized digital media archive for college events.

Dependency on local storage or physical notice boards.

Project Objectives

The primary objectives of this project are:

Centralized Management: To provide a secure Admin Panel for non-technical staff to manage content.

Real-time Updates: To ensure announcements and news are instantly visible to students.

Data Persistence: To move from temporary browser storage (localStorage) to a permanent cloud database (MongoDB).

Resource Accessibility: To allow students to download official notifications as PDF documents.

System Features & Modules

A. Dynamic Notice Board

Categorized announcements (Exams, Admissions, Holidays).

Auto-generated PDF functionality for official documentation.

Filtering system for easy navigation.

B. Faculty & Department Management

Detailed profiles including qualification, experience, and bios.

Department-wise categorization.

Dynamic "Principal's Desk" for vision and mission updates.

C. Media Hub (Digital Archive)

Image gallery for campus facilities.

Video integration for college events and sports.

Search and filter capabilities for media items.

D. Secure Admin Dashboard

Authenticated access to the backend.

CRUD (Create, Read, Update, Delete) operations for all modules.

System overview analytics.

Plan of Work

This project will follow a structured software development process to ensure the portal is fast, secure, and easy to use. The work will be divided into the following steps:

1. Research and Data Collection

I will begin by identifying the specific data needs of GDC Gulabad. This includes gathering sample notices, faculty details, and college media. I will conduct informal interviews with college staff to understand how they currently manage information to ensure the new Admin Panel solves their real-world problems.

2. Development Environment and Instruments

To build the system, I will use the following "instruments" (technical tools):

Visual Studio Code: The primary environment for writing code.

MongoDB Atlas: A cloud-based database to store student and faculty information permanently.

Node.js and Express: To build the logic that handles data requests.

React.js: To create the interactive user interface.

Postman: An instrument used for testing the "handshake" between the frontend and backend.

3. Optimizing Conditions and Testing

To ensure the website works perfectly, I will test it under various conditions:

Responsiveness: Checking the website on mobile phones, tablets, and laptops to make sure the layout looks good on all screens.

Database Speed: Optimizing data queries to ensure that notices and images load quickly, even with a slow internet connection.

Security: Testing the Admin Panel passcode system to prevent unauthorized users from changing college data.